

A.) Transition Matrix P from B' to B :

$$[B|B'] = \left[\begin{array}{cc|cc} 1 & -2 & -12 & -4 \\ 3 & -2 & 0 & 4 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 0 & 6 & 4 \\ 0 & 1 & 9 & 4 \end{array} \right], \text{ so } P = \begin{bmatrix} 6 & 4 \\ 9 & 4 \end{bmatrix}$$

$$B.) [v]_B = P[v]_{B'} = \begin{bmatrix} 6 & 4 \\ 9 & 4 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$[T(v)]_B = A[v]_B = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$C.) \det(P) = (6 \cdot 4) - (4 \cdot 9) = -12$$

$$P^{-1} = -\frac{1}{12} \begin{bmatrix} 4 & -4 \\ -9 & 6 \end{bmatrix} = \begin{bmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{3}{4} & -\frac{1}{2} \end{bmatrix}$$

$$A' = P^{-1}AP = \begin{bmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{3}{4} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 6 & 4 \\ 9 & 4 \end{bmatrix} = \begin{bmatrix} -12 & -\frac{20}{3} \\ \frac{63}{2} & 17 \end{bmatrix}$$

$$D.) [T(v)]_{B'} = P^{-1}[T(v)]_B = \begin{bmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{3}{4} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 \\ -2 \end{bmatrix} = \begin{bmatrix} -\frac{4}{3} \\ \frac{5}{2} \end{bmatrix}$$

$$[T(v)]_{B'} = A'[v]_{B'} = \begin{bmatrix} -12 & -\frac{20}{3} \\ \frac{63}{2} & 17 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -\frac{4}{3} \\ \frac{5}{2} \end{bmatrix}$$